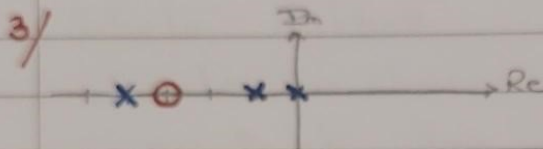


Midterm Spé 2024/2025

1/ ordre = 3 classe = 1

2/ zéros : -3

Pôles : 0 / -1 / -4



4/ $\mathcal{L}(e(t)) = 2/p$

5/ $S(p) = G(p) \cdot E(p) = \frac{2(p+3)}{p^2(p+1)(p+4)}$

6/ $S(p) = \frac{a}{p} + \frac{b}{p^2} + \frac{c}{p+1} + \frac{d}{p+4}$

$$= \frac{ap(p+1)(p+4) + b(p+1)(p+4) + cp^2(p+4) + dp^2(p+1)}{p^2(p+1)(p+4)}$$

$$= \frac{ap^3 + a \cdot 5p^2 + 4ap + bp^2 + b \cdot 5p + 4b + cp^3 + 4cp^2 + dp^3 + dp^2}{p^2(p+1)(p+4)}$$

$$= \frac{(a+c+d)p^3 + (5a+b+4c+d)p^2 + (4a+5b)p + 4b}{p^2(p+1)(p+4)}$$

$$\begin{cases} 4b = 6 \\ 4a + 5b = 2 \\ 5a + b + 4c + d = 0 \\ a + c + d = 0 \end{cases}$$

$$\Rightarrow \begin{cases} b = 3/2 \\ a = \frac{2 - \frac{15}{2}}{4} = -\frac{11}{8} \\ 4c + d = -\frac{3}{2} + \frac{55}{8} = \frac{43}{8} \\ c + d = \frac{11}{8} \end{cases}$$

+/ Si Q6 pas aboutie :

$$S(p) = \frac{a}{p} + \frac{b}{p^2} + \frac{c}{p+1} + \frac{d}{p+4}$$

$$\xrightarrow{\mathcal{L}^{-1}} s(t) = a \cdot u(t) + bt \cdot u(t) + c \cdot e^{-t} u(t) + d e^{-4t} u(t)$$

$$= (a + bt + c e^{-t} + d e^{-4t}) u(t).$$

Exercice 2:

→ Figure 1: VF: 1

$$\text{Dép. relatif} = 0,3 \quad / \quad 30\%$$

$$T_{5\%} \approx 4,5$$

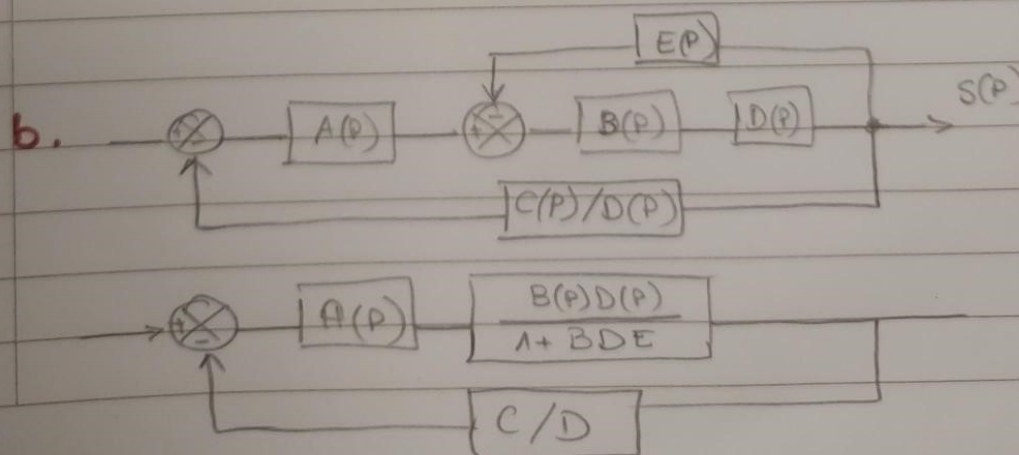
→ Figure 2: VF = 1

$$\text{Dép relatif} = 0,7 \quad / \quad 70\%$$

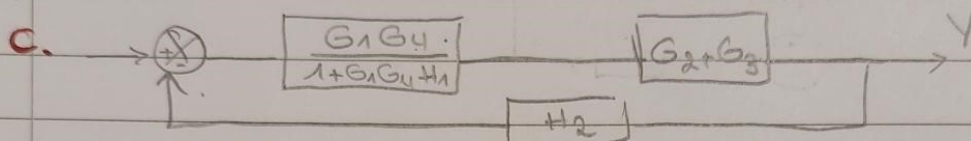
$$T_{5\%} \approx 7,5$$

Exercice 3:

$$a. H(p) = K \cdot \frac{H_1 H_2 H_3}{1 + K H_1 K_2 H_3}$$



$$H(p) = \frac{\frac{ABD}{1+BDE}}{1 + \frac{ABDC}{D(1+BDE)}} = \frac{ABD}{1+BDE+ABC}$$



$$H(p) = \frac{\frac{G_1 G_4 (G_2 + G_3)}{1 + G_1 G_4 H_1}}{1 + \frac{G_1 G_4 (G_2 + G_3) H_2}{1 + G_1 G_4 H_1}} =$$

$$= \frac{G_1 G_4 (G_2 + G_3)}{1 + G_1 G_4 H_1 + G_1 G_4 (G_2 + G_3) H_2}$$

Durée 30 min